ANA515 Project Part 1 - R Script

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```r
#Load necessary libraries
library(dplyr)
library(lubridate)
library(tidyverse)
```

1. EXTRACT

```r
# Load dataset from csv
users_csv  <- read.csv("credit_card_transactions/sd254_users.csv")
cards_csv <- read.csv("credit_card_transactions/sd254_cards.csv")
transactions_csv <- read.csv("credit_card_transactions/credit_card_transactions-ibm_v2.csv")
```

2. TRANFORM

2.1 Modify users table

2.1.a Generate UserId column for users table

2.1.b Rename column names

```r
users <- data.frame(UserId = 1:nrow(users_csv),
                        FullName = users_csv$Person,
                        CurrentAge = users_csv$Current.Age,
                        RetirementAge = users_csv$Retirement.Age,
                        BirthYear = users_csv$Birth.Year,
                        BirthMonth = users_csv$Birth.Month,
                        Gender= users_csv$Gender,
                        Address1 = users_csv$Address,
                        Address2 = users_csv$Apartment,
                        City = users_csv$City,
                        State = users_csv$State,
                        Zipcode = users_csv$Zipcode,
                        Latitude = users_csv$Latitude,
                        Longitude = users_csv$Longitude,
                        PerCapitaIncomeZipcode = users_csv$Per.Capita.Income...Zipcode,
                        YearlyIncome = users_csv$Yearly.Income...Person,
                        TotalDebt = users_csv$Total.Debt,
                        FICOScore = users_csv$FICO.Score,
```

```
                        NumCreditCards = users_csv$Num.Credit.Cards
)

head(users, 3)
```

```
##   UserId       FullName CurrentAge RetirementAge BirthYear BirthMonth Gender
## 1      1 Hazel Robinson         53            66      1966         11 Female
## 2      2     Sasha Sadr         53            68      1966         12 Female
## 3      3     Saanvi Lee         81            67      1938         11 Female
##                 Address1 Address2        City State Zipcode Latitude Longitude
## 1          462 Rose Lane       NA    La Verne    CA   91750    34.15   -117.76
## 2 3606 Federal Boulevard       NA Little Neck    NY   11363    40.76    -73.74
## 3        766 Third Drive       NA West Covina    CA   91792    34.02   -117.89
##   PerCapitaIncomeZipcode YearlyIncome TotalDebt FICOScore NumCreditCards
## 1                 $29278       $59696   $127613       787              5
## 2                 $37891       $77254   $191349       701              5
## 3                 $22681       $33483      $196       698              5
```

2.1.c Convert Gender to categorical data type

```
users$Gender <- factor(users$Gender)
```

2.1.d Convert columns - Address2, Zipcode to Text data type

Zipcode and Address2 should be converted to Text data type because they are not used for quantitative statistics

```
users$Address2 <- as.character(users$Address2)
users$Zipcode <- as.character(users$Zipcode)
```

2.1.e Convert columns - PerCapitaIncomeZipcode, YearlyIncome, TotalDebt to numeric

```
# Remove $ character from the column values and convert to numeric data type
users$PerCapitaIncomeZipcode <- as.numeric(sub('.', '', users$PerCapitaIncomeZipcode))
users$YearlyIncome <- as.numeric(sub('.', '', users$YearlyIncome))
users$TotalDebt <- as.numeric(sub('.', '', users$TotalDebt))
```

2.2 Modify cards tables

2.2.a Remove duplicates

```
cards_csv <- cards_csv %>% distinct()
```

2.2.b Combine user id and card index to generate CardId

2.2.c Rename column names

```
cards <- data.frame(CardId = paste(cards_csv$User, cards_csv$CARD.INDEX, sep = "$"),
                        CardBrand = cards_csv$Card.Brand,
                        CardType = cards_csv$Card.Type,
                        CardNumber = cards_csv$Card.Number,
                        Expires = cards_csv$Expires,
                        CVV = cards_csv$CVV,
                        HasChip = cards_csv$Has.Chip,
                        CardsIssued = cards_csv$Cards.Issued,
                        CreditLimit = cards_csv$Credit.Limit,
                        AcctOpenDate = cards_csv$Acct.Open.Date,
                        YearPINLastChanged = cards_csv$Year.PIN.last.Changed
)
head(cards, 3)
```

```
##   CardId CardBrand CardType   CardNumber Expires CVV HasChip CardsIssued
## 1    0$0      Visa    Debit 4.344677e+15 12/2022 623     YES           2
## 2    0$1      Visa    Debit 4.956966e+15 12/2020 393     YES           2
## 3    0$2      Visa    Debit 4.582313e+15 02/2024 719     YES           2
##   CreditLimit AcctOpenDate YearPINLastChanged
## 1      $24295      09/2002               2008
## 2      $21968      04/2014               2014
## 3      $46414      07/2003               2004
```

2.2.d Convert columns - CardBrand, CardType, HasChip to categorical data type

```
cards$CardBrand <- factor(cards$CardBrand)
cards$CardType <- factor(cards$CardType)
cards$HasChip <- factor(cards$HasChip)
```

2.2.e Convert columns - Expires, AcctOpenDate to Date data type

The Date Format for Expires and AcctOpenDate contains only Month and Year, but R needs a value for Day as well. The Expires date will be set to the last day of the month and AcctOpenDate to the first day of the month for this project.

```
cards$Expires <- my(cards$Expires)
cards$Expires <- ceiling_date(cards$Expires, "month") - days(1)

cards$AcctOpenDate <- my(cards$AcctOpenDate)
```

2.2.f Convert columns - CardNumber, CVV to Text data type

CardNumber and CVV not used for quantitative statistics, therefore they are being converted to Text

```r
cards$CardNumber <- as.character(cards$CardNumber)
cards$CVV <- as.character(cards$CVV)
```

2.2.g Convert column CreditLimit to numeric

```r
cards$CreditLimit <- as.numeric(sub('.', '', cards$CreditLimit))
```

2.3 Create table for merchants

2.3.a Extract Merchant related column from the Transactions table

```r
merchants_temp <- data.frame(MerchantName = transactions_csv$Merchant.Name,
                             MerchantCity = transactions_csv$Merchant.City,
                             MerchantState = transactions_csv$Merchant.State,
                             MerchantZIP = transactions_csv$Zip,
                             MerchantCategoryCode = transactions_csv$MCC
)
```

2.3.b Remove redundant data from the Merchants table

```r
nrow(merchants_temp)
```

```
## [1] 24386900
```

```r
length(unique(merchants_temp$MerchantName))
```

```
## [1] 100343
```

```r
merchants_temp <- merchants_temp %>% distinct(MerchantName, .keep_all = TRUE)
nrow(merchants_temp)
```

```
## [1] 100343
```

2.3.c Generate and add MerchantId column

```r
merchants <- data.frame(MerchantId = 1:nrow(merchants_temp),
                        MerchantName = merchants_temp$MerchantName,
                        MerchantCity = merchants_temp$MerchantCity,
                        MerchantState = merchants_temp$MerchantState,
                        MerchantZIP = merchants_temp$MerchantZIP,
                        MerchantCategoryCode = merchants_temp$MerchantCategoryCode
)

head(merchants, 3)
```

```
##   MerchantId  MerchantName  MerchantCity MerchantState MerchantZIP
## 1          1  3.527213e+18      La Verne            CA       91750
## 2          2 -7.276121e+17 Monterey Park            CA       91754
## 3          3  3.414527e+18 Monterey Park            CA       91754
##   MerchantCategoryCode
## 1                 5300
## 2                 5411
## 3                 5651
```

2.3.d Convert columns - MerchantName, MerchantZIP to Text data type

```r
merchants$MerchantName <- as.character(merchants$MerchantName)
merchants$MerchantZIP <- as.character(merchants$MerchantZIP)
```

2.3.e Convert column MerchantCategoryCode to Categorical data type

```r
merchants$MerchantCategoryCode = factor(merchants$MerchantCategoryCode)
```

2.4 Create table for date

2.4.a Transform Time column to categorical (morning, afternoon, evening, night)

```r
# Convert column to Time data type
transactions_csv$Time <- as.POSIXct(strptime(transactions_csv$Time,"%H:%M"),"UTC")

# Define time range for each category
x=as.POSIXct(strptime(c("050000","105959","110000","155959","160000",
                        "185959"),"%H%M%S"),"UTC")

# Transform Time column
transactions_csv$Time = case_when(
  between(transactions_csv$Time,x[1],x[2]) ~"morning",
  between(transactions_csv$Time,x[3],x[4]) ~"afternoon",
  between(transactions_csv$Time,x[5],x[6]) ~"evening",
  TRUE ~"night")

head(transactions_csv$Time)
```

```
## [1] "morning"   "morning"   "morning"   "evening"   "morning"   "afternoon"
```

2.4.b Extract Date related column from the Transactions table

```r
date_temp <- data.frame(DateYear = transactions_csv$Year,
                        DateMonth = transactions_csv$Month,
                        DateDay = transactions_csv$Day,
                        Time = transactions_csv$Time
)
```

2.4.c Remove redundant data from the Date table

```
date_temp <- date_temp %>% distinct()
```

2.4.d Generate and add DateId column

```
date <- data.frame(DateId = 1:nrow(date_temp),
                        DateYear = date_temp$DateYear,
                        DateMonth = date_temp$DateMonth,
                        DateDay = date_temp$DateDay,
                        Time = date_temp$Time
)
```

2.4.e Convert column Time to Categorical data type

```
date$Time = factor(date$Time)
```

2.5 Fact table - transactions

2.4.a Replace existing merchant columns with single merchant id

```
# Uses Merchant Name (unique column in merchants table)
# to assign Merchant Id values in Transactions table
transactions_csv$MerchantId <- transactions_csv$Merchant.Name
conditions <- merchants$MerchantName
replacement_values <- merchants$MerchantId
transactions_csv$MerchantId <- replace(transactions_csv$MerchantId,
                                    transactions_csv$MerchantId %in% conditions,
                                    replacement_values)
```

2.4.b Replace existing date columns with single date id

```
# Combine all columns in Dates table
temp_dates = data.frame(DateId = date$DateId,
                        Condition = paste(date$DateYear,
                                          date$DateMonth,
                                          date$DateDay,
                                          date$Time)
)

# Combine all columns that were added to Date table from Transaction table
transactions_csv$Condition <- paste(transactions_csv$Year,
                                    transactions_csv$Month,
                                    transactions_csv$Day,
                                    transactions_csv$Time)
```

```r
# Add DateId column with temporary values
transactions_csv$DateId <- 1:nrow(transactions_csv)

# Replace DateId based on the combined Date columns in both tables
transactions_csv$DateId <- replace(transactions_csv$DateID,
                              transactions_csv$Condition %in% temp_dates$Condition,
                              temp_dates$DateId)
```

2.4.c Remove duplicates

```r
transactions_csv <- transactions_csv %>% distinct()
```

2.4.d Combine user id and card index to generate CardId column

2.4.e Rename column names

```r
transactions <- data.frame(TransactionId = 1:nrow(transactions_csv),
                              CardId = paste(transactions_csv$User,
                                             transactions_csv$Card, sep = "$"),
                              UserId = transactions_csv$User,
                              DateId = transactions_csv$DateId,
                              MerchantId = transactions_csv$MerchantId,
                              Amount = transactions_csv$Amount,
                              UseChip = transactions_csv$Use.Chip,
                              Errors = transactions_csv$Errors.,
                              IsFraud = transactions_csv$Is.Fraud
)

head(transactions, 3)
```

```
##   TransactionId CardId UserId DateId MerchantId  Amount           UseChip
## 1             1    0$0      0      1          1 $134.09 Swipe Transaction
## 2             2    0$0      0      2          2  $38.48 Swipe Transaction
## 3             3    0$0      0      3          3 $120.34 Swipe Transaction
##   Errors IsFraud
## 1             No
## 2             No
## 3             No
```

2.4.f Covert columns - UseChip, IsFraud to categorical data type

```r
transactions$UseChip <- factor(transactions$UseChip)
transactions$IsFraud <- factor(transactions$IsFraud)
```

2.2.g Convert column Amount to numeric

```
transactions$Amount <- as.numeric(sub('.', '', transactions$Amount))
```

3. LOAD

Dimension table - users_dim

```
users_dim <- users %>% select(UserId, FullName, CurrentAge, RetirementAge,
                              BirthYear, BirthMonth, Gender, Address1, Address2,
                              City, State, Zipcode, Latitude, Longitude,
                              PerCapitaIncomeZipcode, YearlyIncome, TotalDebt,
                              FICOScore, NumCreditCards) %>% distinct()
summary(users_dim)
```

```
##      UserId         FullName           CurrentAge     RetirementAge
##  Min.   :   1.0   Length:2000        Min.   : 18.00   Min.   :50.00
##  1st Qu.: 500.8   Class :character   1st Qu.: 30.00   1st Qu.:65.00
##  Median :1000.5   Mode  :character   Median : 44.00   Median :66.00
##  Mean   :1000.5                      Mean   : 45.39   Mean   :66.24
##  3rd Qu.:1500.2                      3rd Qu.: 58.00   3rd Qu.:68.00
##  Max.   :2000.0                      Max.   :101.00   Max.   :79.00
##    BirthYear      BirthMonth        Gender       Address1
##  Min.   :1918   Min.   : 1.000   Female:1016   Length:2000
##  1st Qu.:1961   1st Qu.: 3.000   Male  : 984   Class :character
##  Median :1975   Median : 7.000                 Mode  :character
##  Mean   :1974   Mean   : 6.439
##  3rd Qu.:1989   3rd Qu.:10.000
##  Max.   :2002   Max.   :12.000
##    Address2             City              State             Zipcode
##  Length:2000        Length:2000        Length:2000        Length:2000
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##     Latitude       Longitude       PerCapitaIncomeZipcode  YearlyIncome
##  Min.   :20.88   Min.   :-159.41   Min.   :     0         Min.   :     1
##  1st Qu.:33.84   1st Qu.: -97.39   1st Qu.: 16824         1st Qu.: 32818
##  Median :38.25   Median : -86.44   Median : 20581         Median : 40744
##  Mean   :37.39   Mean   : -91.55   Mean   : 23142         Mean   : 45716
##  3rd Qu.:41.20   3rd Qu.: -80.13   3rd Qu.: 26286         3rd Qu.: 52698
##  Max.   :61.20   Max.   : -68.67   Max.   :163145         Max.   :307018
##    TotalDebt        FICOScore     NumCreditCards
##  Min.   :     0   Min.   :480.0   Min.   :1.000
##  1st Qu.: 23987   1st Qu.:681.0   1st Qu.:2.000
##  Median : 58251   Median :711.5   Median :3.000
##  Mean   : 63710   Mean   :709.7   Mean   :3.073
##  3rd Qu.: 89070   3rd Qu.:753.0   3rd Qu.:4.000
##  Max.   :516263   Max.   :850.0   Max.   :9.000
```

Dimension table - cards_dim

```
cards_dim <- cards %>% select(CardId, CardBrand, CardType, CardNumber, Expires,
                              CVV, HasChip, CardsIssued, CreditLimit,
                              AcctOpenDate, YearPINLastChanged) %>% distinct()
summary(cards)
```

```
##     CardId               CardBrand               CardType     CardNumber
##  Length:6146        Amex      : 402   Credit          :2057   Length:6146
##  Class :character   Discover  : 209   Debit           :3511   Class :character
##  Mode  :character   Mastercard:3209   Debit (Prepaid): 578   Mode  :character
##                     Visa      :2326
##
##
##     Expires               CVV            HasChip     CardsIssued
##  Min.   :1997-07-31   Length:6146        NO : 646   Min.   :1.000
##  1st Qu.:2020-02-29   Class :character   YES:5500   1st Qu.:1.000
##  Median :2021-09-30   Mode  :character              Median :1.000
##  Mean   :2020-11-06                                 Mean   :1.503
##  3rd Qu.:2023-05-31                                 3rd Qu.:2.000
##  Max.   :2024-12-31                                 Max.   :3.000
##   CreditLimit      AcctOpenDate        YearPINLastChanged
##  Min.   :     0   Min.   :1991-01-01   Min.   :2002
##  1st Qu.:  7043   1st Qu.:2006-10-01   1st Qu.:2010
##  Median : 12592   Median :2010-02-15   Median :2013
##  Mean   : 14347   Mean   :2011-01-15   Mean   :2013
##  3rd Qu.: 19156   3rd Qu.:2016-05-01   3rd Qu.:2017
##  Max.   :151223   Max.   :2020-02-01   Max.   :2020
```

Dimension table - cards_dim

```
merchants_dim <- merchants %>% select(MerchantId, MerchantName, MerchantCity,
                                      MerchantState, MerchantZIP,
                                      MerchantCategoryCode) %>% distinct()
summary(merchants_dim)
```

```
##    MerchantId     MerchantName       MerchantCity       MerchantState
##  Min.   :     1   Length:100343      Length:100343      Length:100343
##  1st Qu.: 25086   Class :character   Class :character   Class :character
##  Median : 50172   Mode  :character   Mode  :character   Mode  :character
##  Mean   : 50172
##  3rd Qu.: 75258
##  Max.   :100343
##
##  MerchantZIP        MerchantCategoryCode
##  Length:100343      5411   :10051
##  Class :character   5812   : 7910
##  Mode  :character   5912   : 6150
##                     5813   : 5134
##                     5921   : 4701
```

```
##                        4900   : 4695
##                        (Other):61702
```

Dimension table - date_dim

```
date_dim <- date %>% select(DateId, DateYear, DateMonth,
                            DateDay, Time) %>% distinct()
summary(date_dim)
```

```
##      DateId         DateYear      DateMonth         DateDay
##  Min.   :    1   Min.   :1991   Min.   : 1.000   Min.   : 1.00
##  1st Qu.:10366   1st Qu.:1998   1st Qu.: 3.000   1st Qu.: 8.00
##  Median :20731   Median :2005   Median : 7.000   Median :16.00
##  Mean   :20731   Mean   :2005   Mean   : 6.514   Mean   :15.73
##  3rd Qu.:31096   3rd Qu.:2013   3rd Qu.:10.000   3rd Qu.:23.00
##  Max.   :41461   Max.   :2020   Max.   :12.000   Max.   :31.00
##         Time
##  afternoon:10554
##  evening  :10374
##  morning  :10431
##  night    :10102
##
##
```

Fact table - transactions_fact

```
transactions_fact <- transactions %>% select(TransactionId, UserId, CardId,
                                             DateId, Amount, UseChip, Errors,
                                             IsFraud) %>% distinct()
summary(transactions_fact)
```

```
##  TransactionId          UserId        CardId              DateId
##  Min.   :       1   Min.   :   0   Length:24386900    Min.   :    1
##  1st Qu.: 6096726   1st Qu.: 510   Class :character   1st Qu.:10356
##  Median :12193450   Median :1006   Mode  :character   Median :20724
##  Mean   :12193450   Mean   :1001                      Mean   :20726
##  3rd Qu.:18290175   3rd Qu.:1477                      3rd Qu.:31093
##  Max.   :24386900   Max.   :1999                      Max.   :41461
##      Amount                       UseChip            Errors
##  Min.   : -500.00   Chip Transaction  : 6287598   Length:24386900
##  1st Qu.:    9.20   Online Transaction: 2713220   Class :character
##  Median :   30.14   Swipe Transaction :15386082   Mode  :character
##  Mean   :   43.63
##  3rd Qu.:   65.06
##  Max.   :12390.50
##  IsFraud
##  No :24357143
##  Yes:   29757
##
```

```
##
##
##
```